

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 26)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

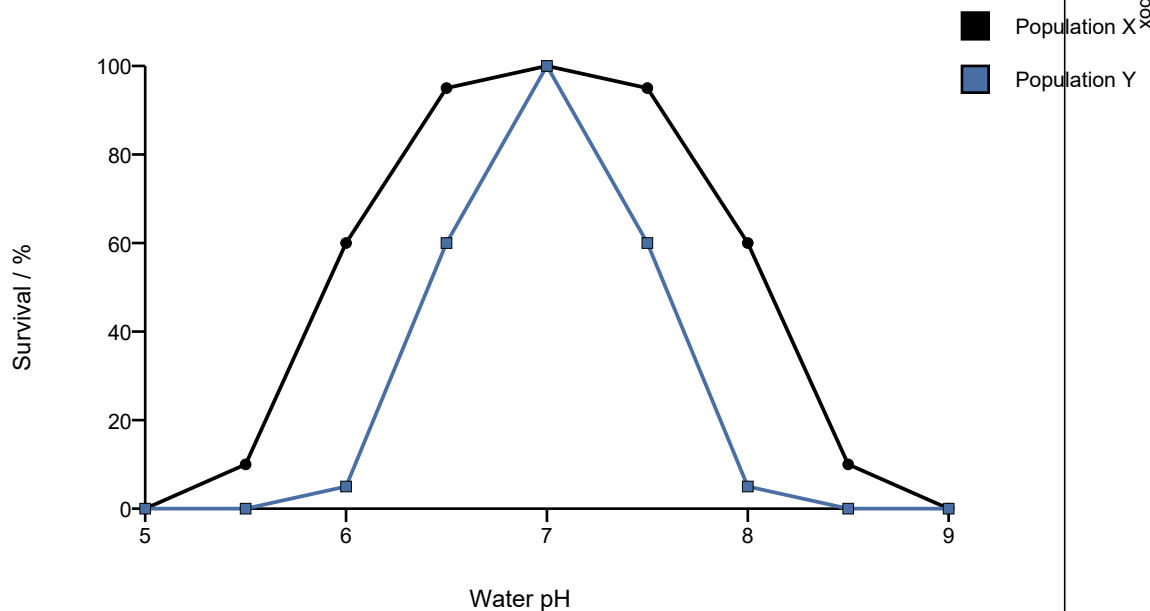
Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 1

. 1

Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

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0 1

. 2

A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

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0 1 . 3

Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

0 1 . 4

Give **two** biotic factors that affect the distribution of a species.

[2 marks]

1

2

10

0 2 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

0 2 . 2

Explain why growing coral reefs help to resist climate change.

[2 marks]

02 . 3

Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

Extra space

15

0	3
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1

 State what is meant by **hybrid vigour**.

[1 mark]

03 . **2** Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

0 3 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

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0 4 . 1

Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 4 . 2

Describe **five** features of a suitable release site.

[5 marks]

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0 4 . 3 Explain the importance of post-release monitoring.

[2 marks]

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10

0 5 . 1 Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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0 5 . 2 Explain how soil organisms provide services that humans depend on.

[3 marks]

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0 5 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

0 5 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

10

0 6

A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

06 . 1

Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

06 . 2

Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using $D = N(N-1) / \sum n(n-1)$. Show your working.

[3 marks]

Answer

06 . 3

The ungrazed meadow had $D = 1.8$ with the same number of species. Explain what this shows.

[2 marks]

06 . 4

Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

06 . 5

Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

10

0 7 Table 1 compares organic and inorganic fertilisers.

0 7 . **1** Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0 7 . **2** Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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10

08 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

- A Close planting of a single non-indigenous species ☐
- B Mixed indigenous species with a mixed age structure ☐
- C Clear-felling on a 40-year rotation ☐
- D Removal of dead wood ☐

08 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

1

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5

09 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

09 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

09 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

5

10

Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

10 . 1

Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer

%

1 0 . 2

Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

1 0 . 3

Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

1 0 . 4

Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

[illegible]

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

OR

1 1 . 2

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Handwriting practice area with 25 horizontal dotted lines.

END OF QUESTIONS